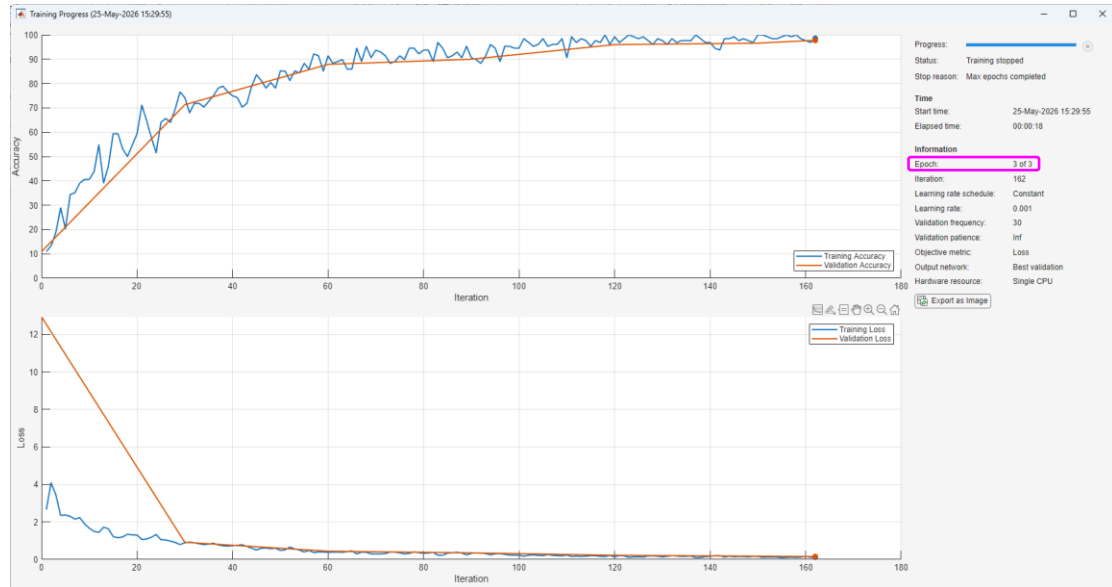


1. Modify the code of

“CreateSimpleDeepLearningClassificationNetworkExample.mlx” according to the following statements and run the modified code.

- Split the data into 70% training and 10% validation, 20% test data.
- Change the solver “sgdm” into “adam,” and name-value pair “MaxEpochs=3.”
- Use the “ValidationData=imdsValidation,” but calculate accuracy = `testnet(net,imdsTest,"accuracy")`.

Example Output:



My result as accuracy = 96.4500

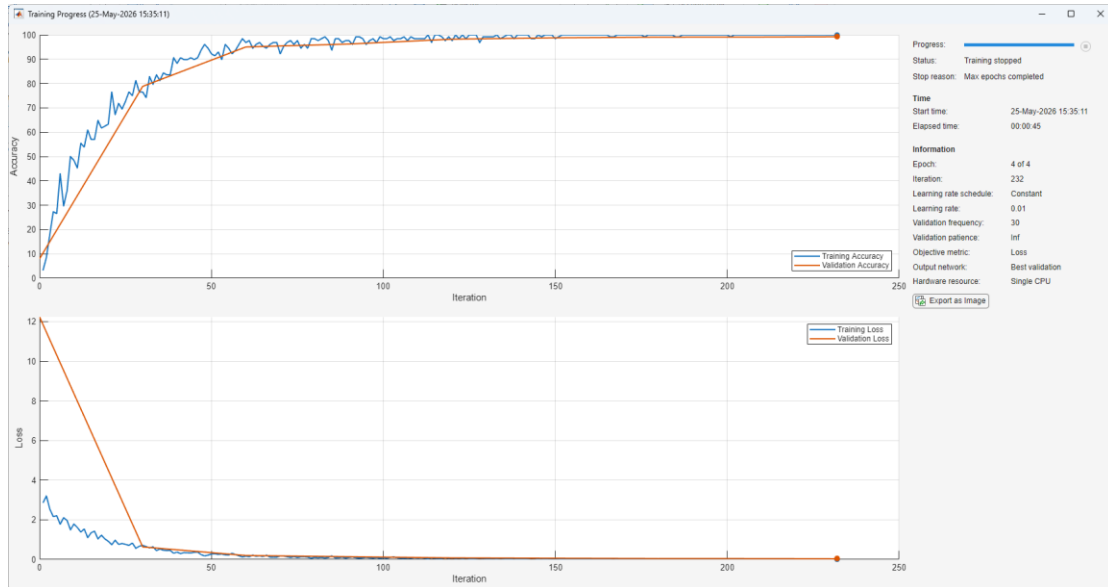
The answer of EX1: submit the file (.mlx) to the system.

2. Modify the code of

“CreateSimpleDeepLearningClassificationNetworkExample.mlx” according to the following statements and run the modified code.

- Replace the “10” in the code “numClasses = 10;” with an automatically calculated value through a built-in function.
- Try to change the contents in the “layers” to make your accuracy higher than 99.0000.

Example Output:



```
accuracy = testnet(net, imdsValidation, "accuracy")
```

```
accuracy = 99.2400
```

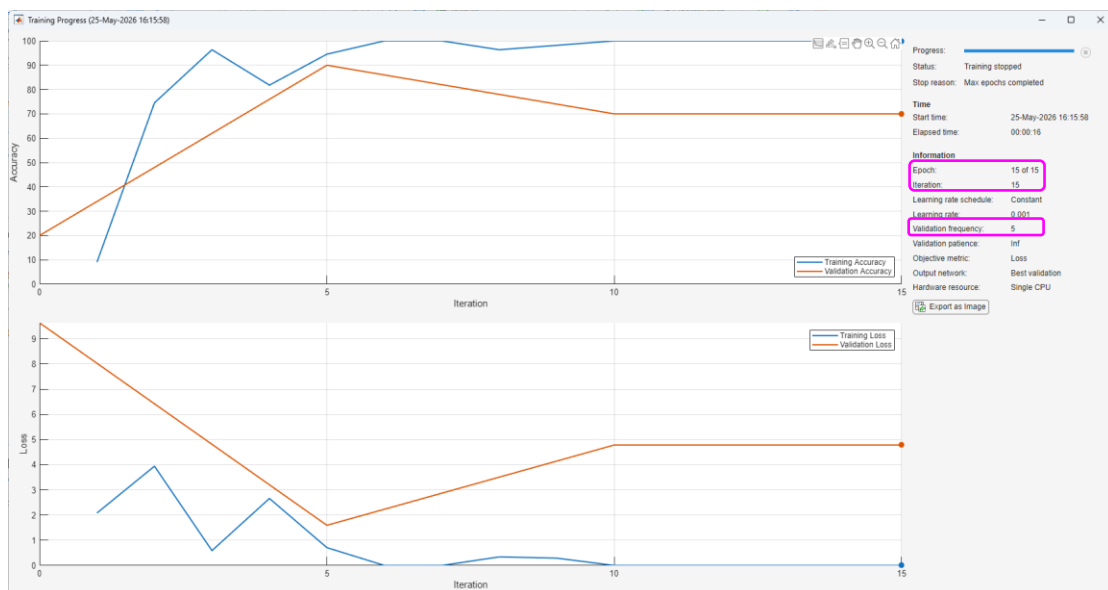
Qualified results.

The answer of EX2: submit the file (.mlx) to the system.

3. Modify the code of

“CreateSimpleDeepLearningClassificationNetworkExample.mlx” to run for “MerchData,” not the original “DigitsData” given inputSize = [227 227 3]. In the trainingOptions, “Solver” as “adam,” “Epoch” as “15” and “Validation frequency” as “5.” In addition, “imdsTrain” occupies 70%, “imdsValidation” 15%, “imdsTest” 15%.

Example Output:



My result is as follows:

```
accuracy = testnet(net, imdsTest, "accuracy")
```

```
accuracy = 80
```

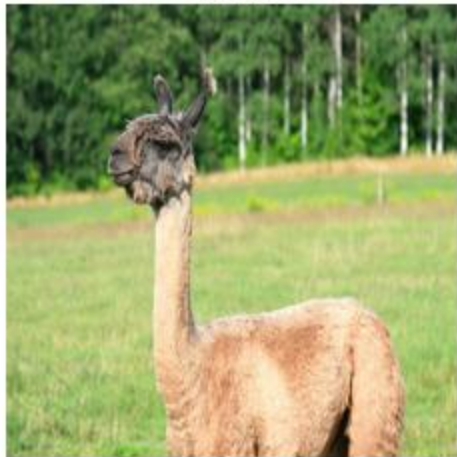
The answer of EX3: submit the file (.mlx) to the system.

4. Modify the code of “ClassifyImageUsingPretrainedNetworkExample.mlx” according to the following statements and run the modified code.
 - a) Change pretrained model “googlenet” as “squeezenet.”
 - b) The original picture “peppers.png” is replaced as “llama.jpg,” “football.jpg,” and “sherlock.jpg” one by one. Therefore, you need to expand some codes to present three recognized pictures simultaneously on the file.

Example Output:

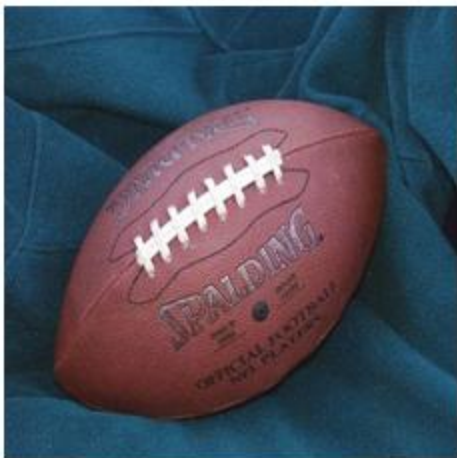
Picture: llama.jpg

llama



Picture: football.jpg

buckle



Picture: sherlock.jpg



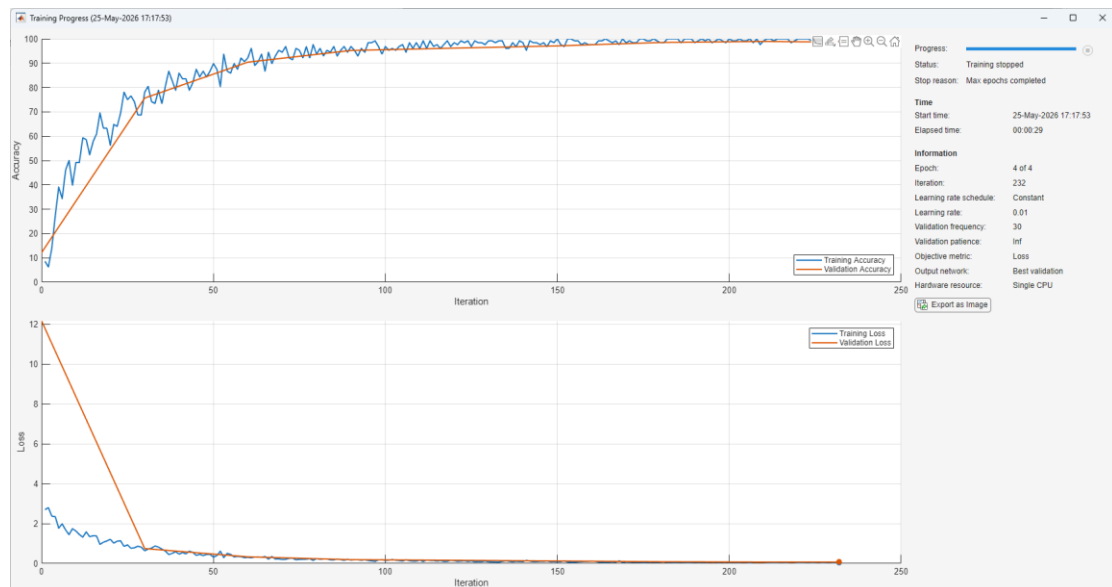
The answer of EX4: submit the file (.mlx) to the system.

5. Modify the code of

“CreateSimpleDeepLearningClassificationNetworkExample.mlx” according to the following statement and run the modified code.

- a) Add part of code from “GetStartedWithTransferLearningExample.mlx” into the previous file to present the confusion matrix.

Example Output:



```
accuracy = testnet(net, imdsValidation, "accuracy")
```

```
accuracy = 98.9600
```

0	250									
1		248	1				1			
2			249						1	
3	1			249						
4					249			1		
5				2	1	245	2			
6	1						247	1	1	
7		3						247		
8	1		2	3			1		241	
9			1						249	
	0	1	2	3	4	5	6	7	8	9

Predicted Class

The answer of EX5: submit the file (.mlx) to the system.